

Sound Analysis Synthesis and Processing: DAAP HW1

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1 Introduction

The talkbox is an acoustic musical effect that routes an instrument signal through a vinyl tube into a performer's mouth, using the oral cavity as a dynamic acoustic filter to shape the sound. This report presents an implementation designed to digitalize this process in MATLAB using Linear Predictive Coding (LPC). The system extracts the time-varying vocal envelope from a speech signal and applies it directly to a musical carrier, simulating the source-filter interaction of the physical instrument.

2 System Architecture and Implementation

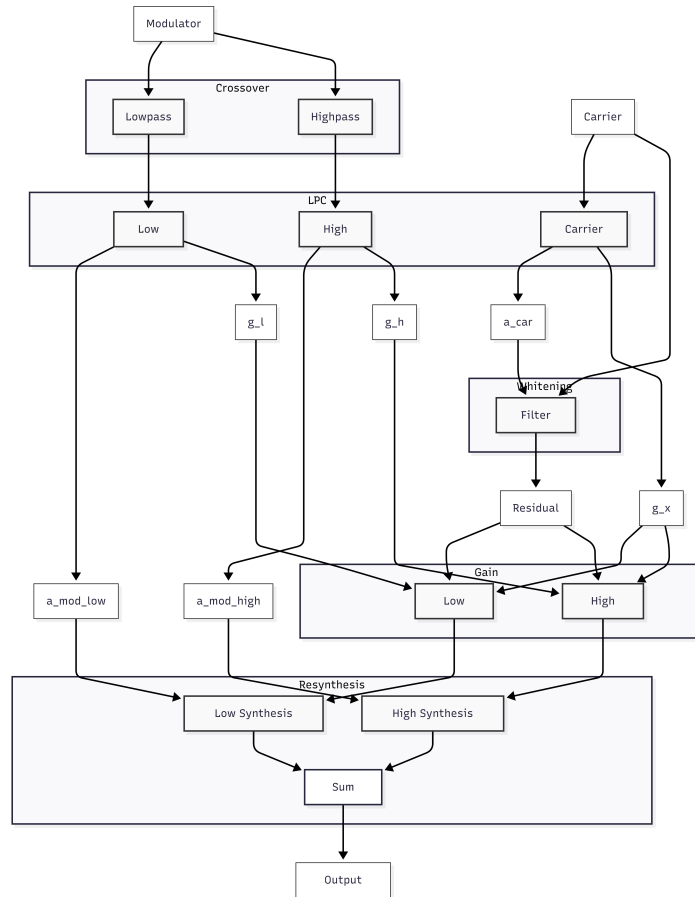


Figure 1: Block diagram of the dual-band LPC talkbox system.

The system architecture and signal flow are illustrated in Figure 1.

First, two audio files, the carrier and modulator signals, are loaded and sent to a preprocessing function to be synchronized in length and sampling rate.

Both signals are processed frame-by-frame using overlap-and-add (OLA) windowing. For each frame, the modulator signal is split at 1 kHz into low and high frequency bands using crossover filters.

The main engine function processes these frames to compute the LPC parameters. The analysis estimates the a_k coefficients that optimize the prediction error, along with their corresponding frame gains.

The resynthesis stage applies these models directly to shape the carrier. The carrier signal is filtered by the inverse whitening filter $A(z)$ to extract the residual (e). This residual is normalized sample-by-sample to match the input envelopes.

The scaled residual drives the recursive all-pole shaping filters ($1/A(z)$) across both bands, which are summed together to construct the synthesized frame. This frame is multiplied by the synthesis window and added to the output vector using OLA to reconstruct the final time-domain output signal.

2.1 Design Choices

The system architecture is designed to be modular, with all configuration parameters at the top of the main script. The window length is adjusted automatically to a power of two to meet the required sampling conditions. To optimize modeling performance, the modulator signal is split into low and high frequency bands. This architectural choice allows the high-frequency consonant components to be modeled using a smaller filter order, which prevents the tracking loop from over-fitting to unvoiced noise.

The implementation relies on two functions to separate preprocessing synchronization from the core LPC engine loop. For feature extraction, the MATLAB `lpc` function is used. Developing and testing a custom LPC solver showed that the built-in function is computationally more efficient and yields better output results. Additionally, a time-delay module was implemented in the output stage to create a stereo widening effect.

2.2 Alternative Experiments

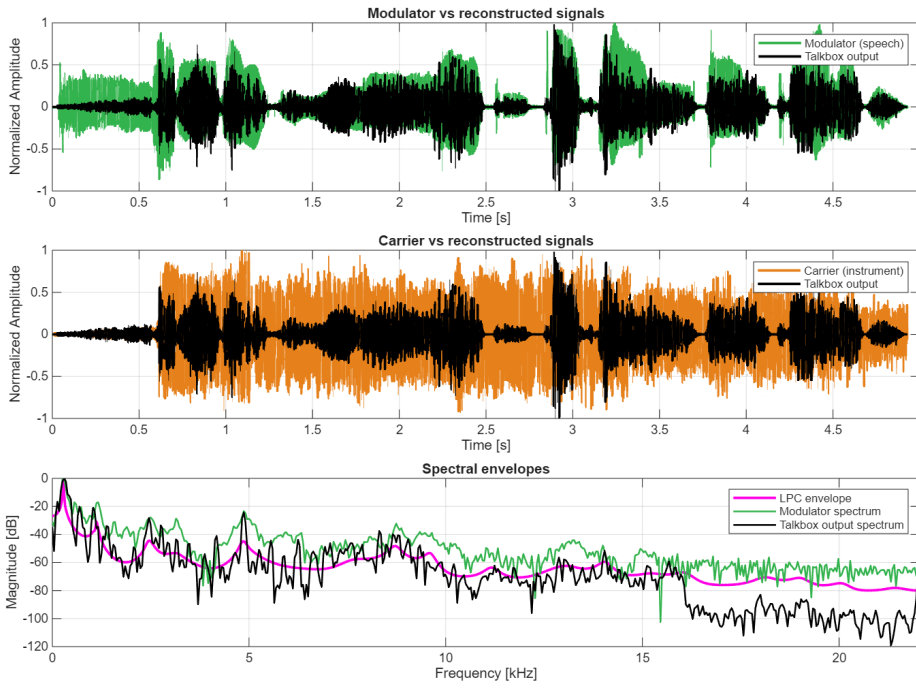
In an alternative version of the system, additional processing stages were implemented to simulate the physical properties of an acoustic talkbox. A low-pass filter was introduced inside the frame loop directly following the resynthesis stage to model the high-frequency dampening caused by the vinyl conduction tube. A compressor was also added at the end of the frame loop to simulate the dynamic saturation of a physical talkbox driver. Finally, a high-pass filter was added just before the final output vector to replicate the acoustic propagation characteristics of the mouth cavity.

Although these additions are theoretically justified by the physics of the system, they significantly increased code complexity without showing clear audio improvements. Furthermore, optimizing the parameters became very complex. For these reasons, the simpler dual-band system was chosen.

3 Experimental Analysis

The following sections present the diagnostic plots for different parameter settings and audio tracks. All figures use the same three-subplot structure :

- **Subplot 1 (Time Domain):** Plots the normalized modulator input and the talkbox output to show temporal tracking and noise gating.
- **Subplot 2 (Time Domain):** Plots the normalized carrier input and the talkbox output to display the amplitude envelope boundaries.
- **Subplot 3 (Spectral Domain):** Displays the middle-frame LPC spectral envelope, the modulator spectrum, and the talkbox output spectrum in decibels across a linear frequency axis.



SYSTEM PARAMETERS

Window length: 23.2 ms = 1024 samples
Hop size: 50% = 512 samples

LPC ORDERS

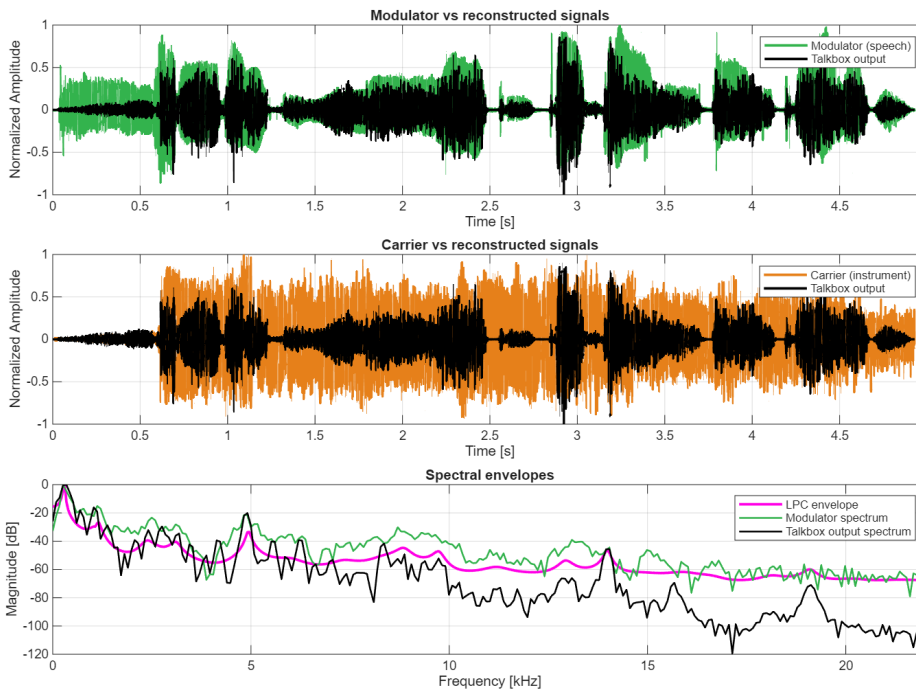
Car: 44, Low: 48, High: 44

Figure 2: System evaluation using the reference files `moore_guitar.mp3` and `Toms_diner.wav`.

In the time domain, the talkbox output successfully follows the modulator speech envelope while remaining generally bounded within the carrier signal amplitude. The output drops to zero during speech pauses, which is coherent with a physical talkbox effect.

In the frequency domain, the LPC envelope tracks the vocal formants, and the talkbox output matches this target shape across the low and mid frequencies.

However, above 15 kHz, the output spectrum falls below the LPC envelope. This might happen because the guitar carrier lacks harmonic energy at such high frequencies, leaving no excitation source to fill out the synthesis filter shape.



SYSTEM PARAMETERS

Window length: 11.6 ms = 512 samples
Hop size: 50% = 256 samples

LPC ORDERS

Car: 44, Low: 48, High: 48

Figure 3: System evaluation using the reference files `moore_guitar.mp3` and `Toms_diner.wav`, with reduced window length.

Reducing the window length to 512 samples improves the temporal reaction to fast speech transients, but it degrades the frequency resolution in the spectral domain, causing the LPC envelope to smooth out closely spaced formant peaks.

AI Usage

An AI tool was utilized during this project to assist with code optimization and report editing.

The tool was introduced only after a working MATLAB script had been independently developed. It was used to improve code efficiency, organize the script into modular functions, and help write comments using accurate terminology.

At each parameter configuration step, the tool helped verify the consistency and mathematical correctness of the results. Finally, AI was used during the writing phase to edit the text to be more accurate and make the explanation more concise.

4 Conclusion

The talkbox engine successfully performs cross-synthesis, allowing vocal formants to shape the instrument carrier signal. However, the synthetic output remains perceptually far from a real physical talkbox. To further develop the system, future work should revisit the alternative filter implementations, or transition to physical modeling techniques to better represent the acoustic of the physical system.